

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding (2)	Algorithm Design (3)	Use of Control Structures (2)	Pseudocode Structure (1)	Completeness (2)	Total(10)
Weightage	20%	30%	20%	10%	20%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:

Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration. (BL3 – 1.05 Marks)

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:

Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. (BL4 – 1.05 Marks)

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Question:

Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. (BL3 – 1.05 Marks)

Q4. Dataset: Student Course Enrollment

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:

Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course "**Python**" are considered during the mining process. Explain how the constraint affects candidate generation and pruning. (BL4 – 1.05 Marks)

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID	Treatments Received
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG



Q4. Constraint-Based Frequent Itemset Mining – "Python" Constraint

Dataset

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Constraint: **Only itemsets containing the course Python are considered.**

Assumed minimum support: 60% = 3 out of 5 students.

The constraint is applied during candidate generation. Instead of creating every possible itemset and later filtering, the mining process starts with Python and only extends candidates that contain Python.

Step 1 – Frequent 1-itemsets under the constraint

Candidate	Support	Decision
{Python}	4/5 = 80%	Keep

Database and AI are not considered as standalone output itemsets because they do not contain Python.

Step 2 – Generate only Python-containing 2-itemsets

Candidate	Transactions	Support	Decision
{Python, Database}	S1,S4	2/5 = 40%	Prune
{Python, AI}	S2,S4	2/5 = 40%	Prune

Since both possible 2-itemsets are infrequent, no larger Python-containing itemset can be frequent under the Apriori property. Thus {Python, Database, AI} is pruned without further support counting.

Constraint-based pseudocode

```
CONSTRAINT-MINING(D, minsup, requiredItem = Python)
1. C1 ← {requiredItem}
2. count support of requiredItem
3. if support(C1) < minsup:
4. return empty set
5. L ← {C1}
6. k ← 2
7. while L(k-1) is not empty:
8. Ck ← generate candidates from L(k-1)
9. remove every candidate that does not contain requiredItem
10. for each candidate c in Ck:
11. if any subset of c is not frequent:
12. prune c
13. count support of remaining candidates
14. Lk ← candidates with support ≥ minsup
15. k ← k + 1
16. return union of all Lk
```

Effect of the constraint: it reduces the search space, prevents irrelevant candidates from being generated, and can reduce computation and memory requirements.

Q5. Hospital Treatment Hierarchy

Visible portion of the supplied dataset

Patient ID	Treatments Received
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG

Important: The uploaded image cuts off after the beginning of Q5. The actual Q5 question, the complete patient list, and the required hierarchy/minimum support information are not visible. Because of that, a precise final Q5 answer would require guessing missing data.

How to complete Q5 once the remaining part is available

- Identify the treatment hierarchy given in the question (for example, Healthcare → Cardiology → ECG, if that is the intended hierarchy).
- Count the treatment occurrences at the required hierarchy level.
- Apply the specified minimum support threshold.
- Generate generalized or specialized frequent itemsets according to the hierarchy constraint.
- Show which candidates are retained and which are pruned.
- If the question asks for pseudocode, provide the hierarchy-aware mining procedure and explain how ancestors/descendants affect support counting.

Please upload the next part/page containing the rest of Q5 if you want the exact Q5 answer included in a revised PDF.

Q3. Association Rule Generation

Given frequent service combinations

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Assumed minimum confidence: 70%.

For an association rule $X \rightarrow Y$, confidence is calculated as:

$$\text{Confidence}(X \rightarrow Y) = \text{Support}(X \cup Y) / \text{Support}(X)$$

Step 1 – Generate rules from {Internet, Streaming}

Rule	Calculation	Confidence	70% filter
Internet $\rightarrow$ Streaming	3/4	75%	Keep
Streaming $\rightarrow$ Internet	3/4	75%	Keep

Step 2 – Generate rules from {Streaming, Cloud Storage}

Rule	Calculation	Confidence	70% filter
Streaming $\rightarrow$ Cloud Storage	2/4	50%	Reject
Cloud Storage $\rightarrow$ Streaming	2/3	66.67%	Reject

Therefore, using 70% as the minimum confidence, the accepted rules are **Internet $\rightarrow$ Streaming** and **Streaming $\rightarrow$ Internet**, each with 75% confidence.

Association-rule pseudocode

```
GENERATE-RULES(FrequentItemsets, minconf)
1. for each frequent itemset I with  $|I| \geq 2$ :
2. generate all non-empty proper subsets X of I
3.  $Y \leftarrow I - X$ 
4. confidence  $\leftarrow$  support(I) / support(X)
5. if confidence  $\geq$  minconf:
6. output rule  $X \rightarrow Y$  with confidence
7. return all accepted rules
```

Filtering can be extended with other measures such as support, lift, leverage or conviction when these values are available. Confidence alone measures how often Y occurs when X occurs; it does not by itself prove that X causes Y.